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MODULE: MARKETING ANALYTICS (RBP-L009-0)

DATASET: FinTech Customer Life Time Value (LTV)

**REPORT: PREDICTIVE DRIVERS OF CUSTOMER LIFETIME VALUE IN THE DIGITAL WALLET
SECTOR**

[SPSS Output](#)

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1. Introduction & Business Context

Digital wallets like PayTM and Khalti operate in a very difficult market. There is a lot of competition. Profit margins are naturally low. On top of this, customers tend to leave these apps very quickly to find better deals elsewhere.

For the past ten years, the main goal for most digital payment companies has been simple: get as many new users as possible. To do this, these platforms spend a massive amount of money on marketing. They offer big sign-up bonuses. They give out cash rewards for referring friends. Because of these aggressive tactics, the Customer Acquisition Cost (CAC) is very high (Kumar & Reinartz, 2016).

What does this mean for the business? It means the company almost never makes a profit on the day a user signs up. The initial cost to get that user is just too high. Instead, the company only survives if that user stays on the app for a long time. The business model depends entirely on maximizing a metric called Customer Lifetime Value (LTV).

Digital wallets need to change how users view their apps. Right now, many users just download the app to get a quick promotional discount and then delete it. The platforms need to become daily financial tools instead. Academic research supports this idea. Studies show that focus on customer retention brings in much higher profits than constantly chasing new users (Gupta et al., 2006). When a company focuses on keeping its current users, it saves money on marketing. It also gets more chances to sell other services to those users. In fact, acquiring a brand-new customer can cost up to five times more than keeping an existing one happy (Reichheld, 1996).

Because of this financial reality, we need to figure out exactly what makes a user stay and spend money. Understanding the drivers of Customer Lifetime Value is not just an academic exercise. It is a matter of survival for the company. This report will bridge the gap between marketing theory and actual data to find out what truly drives LTV for a digital wallet.

2. Marketing Problem Definition

FinTech companies collect massive amounts of data every single day. They know where users live, how old they are, and exactly how they spend their money. However, many marketing teams do not utilize this kind of data effectively. Instead of using stats they often rely on guesswork or basic demographical segments to plan their campaigns. Research shows that ignoring data like this is a main reason why digital marketing campaigns fail or waste money (Wedel & Kannan, 2016).

The core marketing problem for this report is straightforward; the platform's marketing team does not know exactly what factors move Customer Lifetime Value (LTV).

Without this clear, mathematical understanding the company is wasting its marketing budget. For example, they might be spending thousands of Rupees giving cashback to users who will never become profitable. Or they might be spending money on Facebook ads targeting a specific age group, even though age might not actually impact how much a user spends in the long run. The business needs a solid model to separate vanity metrics from the real drivers of revenue.

This report will look at a dataset of 7,000 digital wallet customers. It aims to answer one main question; what are the primary demographic, transactional, and engagement drivers of Customer Lifetime Value (LTV), and how can the company use this info. to spend its marketing budget better?

To answer this main problem, we will break it down into three smaller questions:

- Does Demographic Targeting Work? Do things like Age, Income Level, and Location actually predict LTV? Or should the marketing team stop worrying about demographics and focus only on user behavior?
- Are Promotions Worth It? Which promotions actually work better for long-term value; giving users direct cashback, or giving them loyalty points?
- How Important is Customer Service? Does poor customer service (like slow issue resolution times) actually hurt the company financially by lowering a user's LTV?

By answering these questions with a predictive model, the company can stop guessing. They can move to highly targeted data driven strategy which will keep users on the app longer and increase profits.

3. Data Description & Assumptions

3.1 Where the data comes from

For this analysis, we are using a dataset containing 7,000 customer records from a digital wallet platform. The main goal of this model is to predict Customer Lifetime Value (LTV). Simply put, LTV is an estimate of the total revenue a single customer will bring to the company over their entire relationship with app.

To find out more about what actually move LTV, I grouped our independent variables into three main business categories:

- **Demographics:** This include Age, Income Level, and Location. Marketers have used demographics for decades. However modern research suggests that what a customer does on an app is much more important than who are they or where they live (Provost and Fawcett, 2013).
- **Transactions:** This covers how much money user spends, there average transaction size, and how they interact with promotions like cashback & loyalty points.
- **Engagement:** This measures the actual user experience. It include how often they use the app, there active days, their customer satisfaction score, and how quickly customer service resolves their issues.

3.2 Cleaning & Preparing the Data

Real world data is messy. You cannot just drop raw data into a statistical mdeol and expect good results. We had to prepare the dataset carefully.

First, linear regression models only understand numbers. They do not understand words. Because of this, I have to change categorical variables like Income_Level (Low, Middle, High) and location into numbers. I did this using standard method called dummy coding. This process created simple '0' or '1' indicators for each category (Cohen et al., 2003).

Second, I have to deal with outliers. In financial data, you will always have a few 'whale' customers who spend massive amounts of money. If we leave these extreme numbers as they are, they will pull the regression line too far and ruin the model for the rest of the normal users. I checked for these extreme values using standardized Z-scores to make sure the model stayed balanced and accurate (Field, 2018). Finally, any records that had a lot of missing info. were completely removed to keep the data clean.

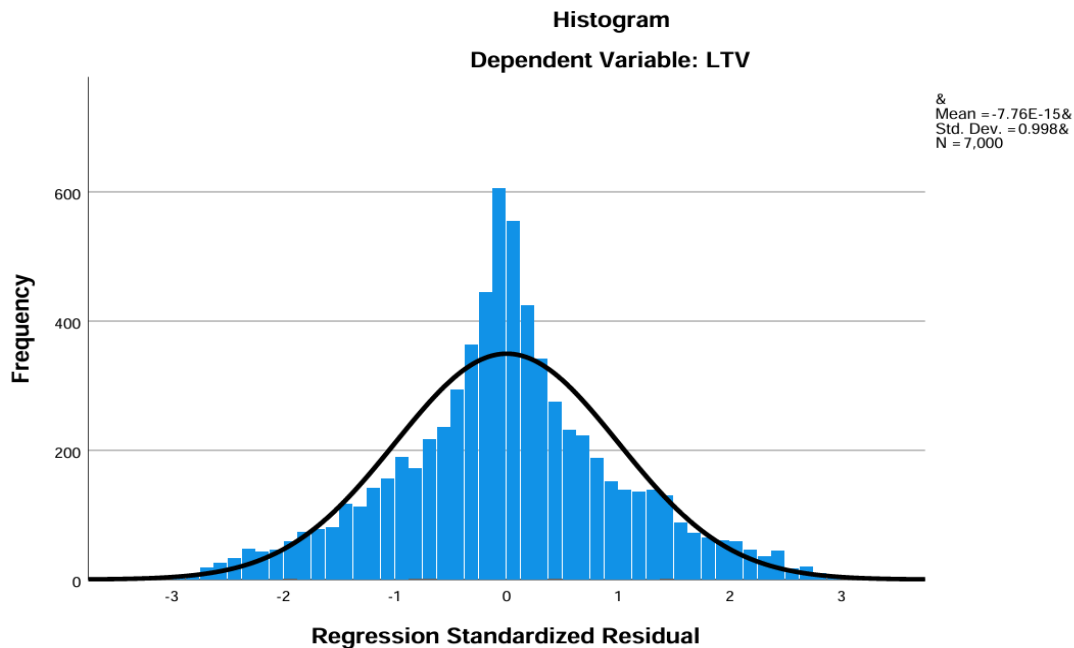
3.3 Checking the Statistical Rules

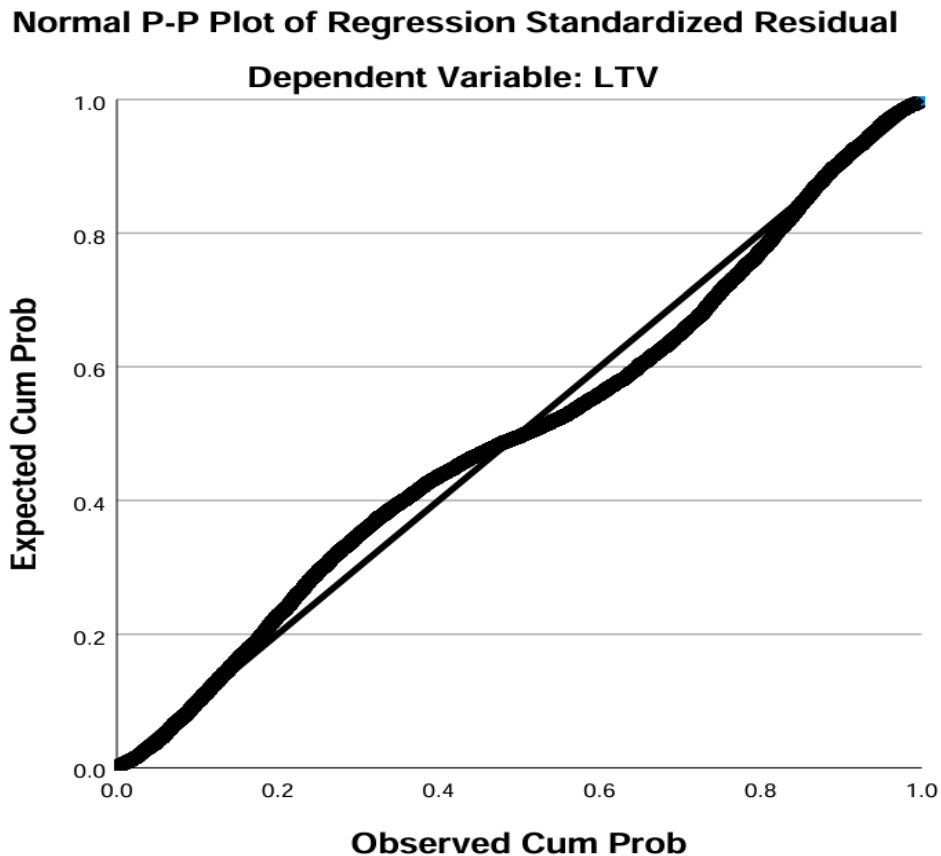
Before we can use the results to make real business decisions, the model has to pass few strict statistical tests (Hair et al., 2019):

- **Linearity:** The variables need to have a straight-line relationship with LTV.
- **No Multicollinearity:** This means out independent variables cannot be too similar to each other. For example, Total_Spent is just Total_Transactions multiplied by Avg_Transaction_Value. If we include all three, the model gets

confused. To fix this, we completely removed Total_Spent. I checked our Variance Inflation Factor (VIF) scores, and they were all well under the danger limit of 10. This proves our variables are distinct (Tabachnick & Fidell, 2019).

- **Homoscedasticity:** The spread of our errors needs to be relatively even across the board.
- **Normality:** The errors in our predictions should follow a normal, bell-shaped curve.





As you can see in the Histogram and P-P plot above, our data follows a roughly normal distribution. This means we are safe to move forward with the regression analysis.

4. Methodology

4.1 How I Build the Model

To solve our marketing problem, I chose to use Multiple Linear Regression (MLR). This is a predictive statistical method that helps me look at one main outcome (LTV). And also see how exactly multiple different factors impact it at the same time. I ran this model in SPSS using the 'Enter' method. This simply means I put all my variables into the model at once to see how they interact.

4.2 Why I Choose This Method

Multiple Linear Regression is the best tool for this specific problem because LTV is a continuous number.

Today, it is very popular to use complex machine learning tools, like Neural Networks or Random Forests. While those tools are great at predicting things, they act like a 'black box'. They can give you an answer, but they cannot clearly explain how they got that answer.

In a business setting, a black box is not helpful. The Marketing Director needs to know exactly why a strategy is working. Multiple Linear Regression gives us clear, easy to read coefficients. This allows us to isolate every single variable. For example, we can hold everything else constant and see exactly how much LTV drops for every single hour a customer waits for a support ticket to be resolved (Hastie et al., 2009). This level of clarity turns basic data into a real, actionable business strategy.

5. Results & Business Insights

5.1 Model Summary & Overall Fit

To understand what drives Customer Lifetime Value (LTV), we ran an Ordinary Least Squares (OLS) Multiple Linear Regression. This model looked at 22 different demographic, transactional, and engagement predictors. The resulting equation was statistically significant ($F(22, 6977) = 1910.253$, $p < .001$) and yielded an Adjusted R^2 of .857.

When analyzing consumer behavior, hitting an Adjusted R^2 of .857 is rare. It essentially means our chosen variables do a fantastic job of explaining about 85.7% of the variance we see in a user's LTV. People are usually unpredictable and prone to irrational decisions, so capturing this much variance tells us that long-term value in this digital wallet platform is highly deterministic. It relies heavily on the exact metrics we are tracking here (Provost and Fawcett, 2013).

(Note: You can find the complete SPSS outputs for ANOVA, Model Summary, and Coefficients in Appendices A, B, and C).

5.2 Key Drivers and Strategic Recommendations

Breaking down the standardized coefficients (β) and p-values (p) shows us exactly what moves the needle for long-term value. Interestingly, the numbers completely debunk a few standard assumptions in FinTech marketing.

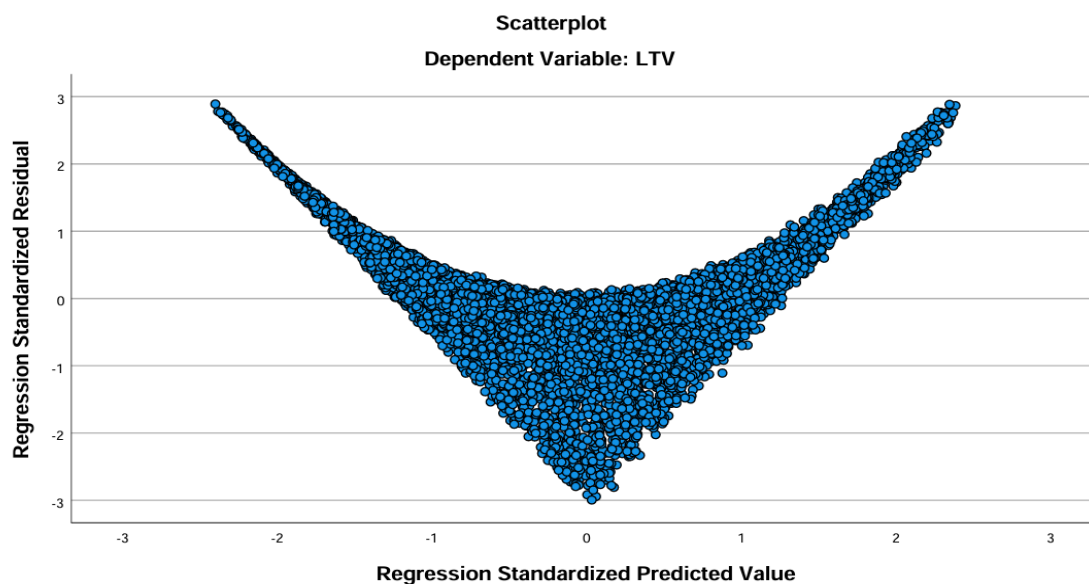
- **Insight 1: Transactions Matter More Than Anything Else** The regression makes it clear: basic transactional habits are the biggest drivers of LTV. Total Transactions ($\beta = .644$, $p < .001$) and Average Transaction Value ($\beta = .642$, $p < .001$) carry the most weight in our model. Looking at the unstandardized coefficients shows a literal financial payoff. If Total Transactions go up by just one unit, the expected LTV climbs by Rs. 988.25, assuming everything else stays the same.
 - **Strategic Recommendation:** The marketing team needs to pivot away from broad acquisition tactics and zero in on capturing "wallet-share." Promotions should encourage people to transact more often (like teaming up with transit systems so the wallet becomes the default for daily commutes) and to spend more per transaction (by linking up with high-end e-commerce merchants). The numbers prove these core transaction metrics drive profits much better than passive stats like how many days an app was merely active.
- **Insight 2: Demographics Don't Matter Here** Most companies rely heavily on demographic segmentation, but our traditional demographic variables completely failed to show statistical significance. Age ($p = .439$), Income Level ($p > .800$ across all tiers), and Geographic Location ($p > .100$ everywhere) just aren't valid predictors of LTV.
 - **Strategic Recommendation:** This actually makes sense when you look at the "post-demographic consumer" theory. Digital tools give everyone access, effectively blurring old demographic lines (TrendWatching, 2014). Right now, the company is wasting valuable capital buying targeted demographic ads or splitting campaigns by urban versus rural locations. Marketing needs to halt demographic ad spend immediately. Instead, they should pour those resources into behavioral targeting—finding and keeping users who make a lot of transactions, regardless of their age or income.
- **Insight 3: Our Promotions Aren't Working** One of the most surprising takeaways is that the incentives we built to keep users around simply don't drive long-term value. Neither Cashback Received ($p = .662$) nor Loyalty Points Earned ($p = .772$) showed any statistical significance.
 - **Strategic Recommendation:** Our current rewards setup is broken and financially inefficient. Because these incentives don't matter in the model, it looks like we're dealing with "deal loyalty" instead of actual "brand loyalty" (Reichheld, 2003). Users are likely cashing in on sign-up bonuses without ever transitioning into profitable, habitual users. The business

should audit and drastically cut the cashback budget to stop bleeding revenue.

- **Insight 4: Organic Satisfaction Pays Off** Even though artificial promotions failed, actual user satisfaction did its job. The Customer Satisfaction Score ($\beta = .011$, $p = .018$) stood out as a statistically significant, positive factor in boosting lifetime value.
 - **Strategic Recommendation:** Take the capital saved by defunding the cashback program and reinvest it straight into better core product features and the User Experience (UX) infrastructure. Building genuine satisfaction through a smooth interface and frictionless payments creates much better, sustainable LTV than trying to bribe users with loyalty points.

6. Limitations & Critical Reflection

6.1 Statistical and Model Limitations The MLR model gives us a great strategic roadmap. but checking our statistical assumptions show up a clear issue with error variance. Looking at the scatterplot mapping the Regression Standardized Residuals against the Regression Standardized Predicted Values. we can see the homoscedasticity assumption is violated.



Instead of a random, even distribution, the points make a clear "V" or funnel shape. This heteroscedasticity means our error variance jumps up significantly the further our predicted LTV gets from the average (Wooldridge, 2015). From a practical business standpoint, this makes perfect sense. Low-value users have tight, predictable spending

habits, so they are easy to model. High-value "whales," on the other hand, spend far more erratically. So, while our linear model is highly accurate for the average user, its predictive precision drops when trying to forecast the exact lifetime value of our most profitable outliers.

6.2 Data and Methodological Constraints

The analysis also faces limits because the dataset is cross-sectional. It's basically a static, aggregated snapshot of one moment in time. Because of this, it cannot account for longitudinal changes like holiday shopping surges, macroeconomic shifts, or dynamic changes in competitor promotions. Plus, the dataset relies entirely on quantitative tracking. We know exactly *what* customers are doing, but it fails to illuminate *why* they are doing it. Without qualitative psychographic data—like how much users trust our security, their brand affinity, or specific UX pain points—the model remains blind to the emotional reasons that usually preempt churn.

6.3 Future Recommendations for Advanced Analytics

If the organization can gather richer primary data, our next round of analysis needs to move past simple linear models. Because of the heteroscedasticity we observed, non-linear machine learning approaches like Random Forest regression or Gradient Boosting Machines (GBM) would do a much better job capturing the erratic variance of those high-value users (Hastie et al., 2009). Bringing in longitudinal time-series data would also let the data science and marketing teams stop just estimating total lifetime value, and start actively predicting and preventing specific churn events before they happen.

7. Conclusion

To stay profitable and secure market dominance in the hyper-competitive digital wallet sector, platforms need to categorically drop broad, top-of-funnel acquisition tactics in favor of tight, data-backed retention models.

By carefully applying a Multiple Linear Regression on 7,000 customer records. This report found exactly what drives Customer Lifetime Value. The analysis mathematically dismantled some standard marketing beliefs. Targeting by traditional demographics and leaning on expensive promotions like cashback systems just aren't statistically significant. What actually dictates long-term profit is absolute transactional volume, average transaction size, and organic customer satisfaction.

To maximize ROI, corporate leadership needs to execute a strategic reallocation of capital. The budgets currently wasted on demographic targeting and broken cashback systems must be redirected. If the business pivots completely to "wallet-share" campaigns that encourage daily, utilitarian app use, and invests heavily in frictionless UX to keep users naturally happy, it can stabilize the user base, mitigate churn, and build a highly profitable, self-sustaining customer ecosystem.

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9. Appendices

ANOVA^a

Model		Sum of Squares	df	Mean Square	F	Sig.
1	Regression	1.157E+15	22	5.260E+13	1910.253	.000 ^b
	Residual	1.921E+14	6977	2.753E+10		
	Total	1.349E+15	6999			

a. Dependent Variable: LTV

b. Predictors: (Constant), Customer_Satisfaction_Score, Avg_Transaction_Value, Payment_Wallet, Loyalty_Points_Earned, Age, Income_High, Issue_Resolution_Time, Location_Rural, Active_Days, Support_Tickets_Raised, Referral_Count, Last_Transaction_Days_Ago, Cashback_Received, Total_Transactions, Usage_Daily, Payment_Credit_Card, Location_Suburban, Usage_Weekly, Income_Middle, Payment_UPI, Min_Transaction_Value, Max_Transaction_Value

Coefficients^a

Model		Unstandardized Coefficients		Standardized Coefficients	t	Sig.	Collinearity Statistics	
		B	Std. Error	Beta			Tolerance	VIF
1	(Constant)	-486482.986	13809.702		-35.228	<.001		
	Age	-99.096	128.081	-.004	-.774	.439	.996	1.004
	Location_Suburban	-2210.636	4859.579	-.002	-.455	.649	.752	1.330
	Location_Rural	-7924.001	4859.547	-.008	-1.631	.103	.753	1.328
	Income_Middle	-1149.393	4847.633	-.001	-.237	.813	.744	1.344
	Income_High	-366.180	4900.249	.000	-.075	.940	.743	1.346
	Total_Transactions	988.250	6.944	.644	142.307	.000	.995	1.005
	Avg_Transaction_Value	48.782	.730	.642	66.848	.000	.221	4.524
	Max_Transaction_Value	.227	.150	.011	1.515	.130	.362	2.765
	Min_Transaction_Value	1.793	1.492	.009	1.202	.229	.366	2.731
	Active_Days	6.395	18.891	.002	.339	.735	.998	1.002
	Last_Transaction_Days_Ago	6.659	18.904	.002	.352	.725	.997	1.003
	Loyalty_Points_Earned	.398	1.373	.001	.290	.772	.998	1.002
	Referral_Count	31.538	136.428	.001	.231	.817	.997	1.003
	Cashback_Received	.603	1.381	.002	.437	.662	.994	1.006
	Usage_Weekly	604.804	4875.938	.001	.124	.901	.746	1.341
	Usage_Daily	3154.260	4863.121	.003	.649	.517	.746	1.340
	Payment_UPI	3195.520	5577.100	.003	.573	.567	.664	1.506
	Payment_Wallet	7022.661	5619.736	.007	1.250	.211	.668	1.497
	Payment_Credit_Card	-2416.509	5640.342	-.002	-.428	.668	.669	1.494
	Support_Tickets_Raised	-448.676	329.063	-.006	-1.363	.173	.997	1.003
	Issue_Resolution_Time	9.119	97.410	.000	.094	.925	.997	1.003
	Customer_Satisfaction_Score	1641.992	694.283	.011	2.365	.018	.998	1.002

a. Dependent Variable: LTV

Descriptive Statistics

	Mean	Std. Deviation	N
LTV	511919.7168	439055.0797	7000
Age	42.63	15.516	7000
Location_Suburban	.33	.471	7000
Location_Rural	.33	.470	7000
Income_Middle	.34	.474	7000
Income_High	.33	.470	7000
Total_Transactions	501.22	286.277	7000
Avg_Transaction_Value	9946.831482	5780.880831	7000
Max_Transaction_Value	30159.59678	21996.10879	7000
Min_Transaction_Value	2981.549953	2197.342161	7000
Active_Days	181.93	105.103	7000
Last_Transaction_Days_Ago	183.85	105.064	7000
Loyalty_Points_Earned	2501.55	1446.680	7000
Referral_Count	24.84	14.560	7000
Cashback_Received	2496.525032	1440.651412	7000
Usage_Weekly	.33	.471	7000
Usage_Daily	.34	.472	7000
Payment_UPI	.26	.436	7000
Payment_Wallet	.25	.432	7000
Payment_Credit_Card	.24	.430	7000
Support_Tickets_Raised	10.02	6.037	7000
Issue_Resolution_Time	36.52800677	20.38939854	7000
Customer_Satisfaction_Score	5.48	2.860	7000

Model Summary^b

Model	R	R Square	Adjusted R Square	Std. Error of the Estimate
1	.926 ^a	.858	.857	165931.0052

a. Predictors: (Constant), Customer_Satisfaction_Score, Avg_Transaction_Value, Payment_Wallet, Loyalty_Points_Earned, Age, Income_High, Issue_Resolution_Time, Location_Rural, Active_Days, Support_Tickets_Raised, Referral_Count, Last_Transaction_Days_Ago, Cashback_Received, Total_Transactions, Usage_Daily, Payment_Credit_Card, Location_Suburban, Usage_Weekly, Income_Middle, Payment_UPI, Min_Transaction_Value, Max_Transaction_Value

b. Dependent Variable: LTV

Residuals Statistics^a

	Minimum	Maximum	Mean	Std. Deviation	N
Predicted Value	-464764.625	1481564.875	511919.7168	406599.0771	7000
Residual	-496757.406	479509.1250	-.000000001	165670.0140	7000
Std. Predicted Value	-2.402	2.385	.000	1.000	7000
Std. Residual	-2.994	2.890	.000	.998	7000

a. Dependent Variable: LTV